Chapter S

Additional files

This document holds the tables and figures that support the main chapters but are too long to include in them. They are grouped by the chapter they support and numbered in order as Additional files.

S.1 Noise robustness across molecular representations and model architectures

Additional file 1: Hyperparameters for every model configuration, and the descriptors that make up PDV

Table A lists the default settings of every model configuration, Table B the tuned settings that replace a default on each dataset, and Table C the 200 descriptors that make up PDV.

Table S.1: Table A: default hyperparameters for every model configuration in the study. Every value is read from `models/model_defaults.py` at spec version 1.7.0, spec hash `9249926c38dc`, which is the file both the QM9 pipeline and the assay pipeline load their settings from. Every QM9 results row carries the spec version and the spec hash as two of its columns, so a QM9 table can be traced back to the settings that produced it; the three assay datasets are written by a second pipeline whose results files do not carry those two columns. The configuration list is the model roster the job generator submits. One setting is used for every representation and for every dataset, except where Table B gives a tuned replacement. The support vector machine uses an RBF kernel on every representation, so no result in this study carries a kernel that changes with the representation. Two values are not held in the shared spec and name their source file in the third column.

Configuration	Hyperparameter	Default value
Shared by every neural network below		

Configuration	Hyperparameter	Default value
Training, all networks	optimizer	adam
	lr	0.001
	weight_decay	0.0
	batch_size	32
	epochs	100 cap; early stopping usually reaches it first
	patience	10 epochs without a strictly better mean validation loss
	restore_best_weights	True the reported fit is the best epoch, not the last
	val_loss_reduction	mean
	improvement_tolerance	0.0
	mc_passes	100 stochastic forward passes at inference
	standardise_targets	True fitted on training labels alone
Tree and boosting models		
RF `rf`	n_estimators	100
	max_depth	None no limit
	min_samples_leaf	5
	min_samples_split	2
	max_features	0.3
	bootstrap	True
RF300 `rf300`	n_estimators	300
	max_depth	None no limit
	min_samples_leaf	5
	min_samples_split	2
	max_features	0.3
	bootstrap	True
XGBoost `xgboost`	n_estimators	100
	max_depth	6
	learning_rate	0.1
	subsample	1.0
	colsample_bytree	1.0
	colsample_bylevel	1.0
	min_child_weight	1
	gamma	0.0

Configuration	Hyperparameter	Default value
	reg_alpha	0.0
	reg_lambda	1.0
LightGBM `lgb`	n_estimators	100
	learning_rate	0.1
	num_leaves	31
	max_depth	−1 no limit
	subsample	1.0
	colsample_bytree	1.0
	min_child_samples	20
	reg_alpha	0.0
	reg_lambda	0.0
NGBoost `ngboost`	n_estimators	500
	learning_rate	0.01
	natural_gradient	True
	dist	Normal ngboost.distns.Normal
	score	MLE ngboost.scores.MLE, the log scoring rule
	minibatch_frac	1.0
	col_sample	1.0
	base_max_depth	3 depth of one boosting stage
	base_criterion	friedman_mse
	early_stopping_rounds	50 stages without a better validation log likelihood
	use_best_iteration	True predictions use the best stage, not every stage fitted
	ngboost_ensemble_seeds	1 1 = one fit, no seed ensemble
QRF `qrf`	n_estimators	300
	max_depth	None no limit
	min_samples_leaf	5
	min_samples_split	2
	max_features	0.3
	bootstrap	True
Kernel models		

Configuration	Hyperparameter	Default value
SVM `svm`	C	1.0
	gamma	scale
	kernel	rbf
GP `gauche_rbf`	kernel	rbf set by this job, not by the shared spec
	likelihood_noise	0.001 starting value; it is learned
	outputscale	1.0
	apply_outputscale	True
	max_train_n	5000 molecules the process is fitted on; an exact Gaussian process is cubic in this. The test set is untouched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1 used only when the standard fitter refuses the model
	fallback_adam_iters	100 used only when the standard fitter refuses the model
	single_thread_fit	False a threading workaround, off since the environment check found one runtime
	init_lengthscale_from_data	True the RBF length scale starts at the median distance between training molecules, not at the library default
	lengthscale_probe_n	500 molecules sampled to estimate that median

Configuration	Hyperparameter	Default value
	collapse_fraction	0.05 a fit whose predictions vary by less than this fraction of the training label spread is recorded as collapsed rather than scored
GP-Tanimoto `gauche` run on ecfp4 only	kernel	tanimoto set by this job, not by the shared spec
	likelihood_noise	0.001 starting value; it is learned
	outputscale	1.0
	apply_outputscale	True
	max_train_n	5000 molecules the process is fitted on; an exact Gaussian process is cubic in this. The test set is untouched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1 used only when the standard fitter refuses the model
	fallback_adam_iters	100 used only when the standard fitter refuses the model
	single_thread_fit	False a threading workaround, off since the environment check found one runtime
	init_lengthscale_from_data	True the RBF length scale starts at the median distance between training molecules, not at the library default
	lengthscale_probe_n	500 molecules sampled to estimate that median

Configuration	Hyperparameter	Default value
	collapse_fraction	0.05 a fit whose predictions vary by less than this fraction of the training label spread is recorded as collapsed rather than scored
GP-Hetero `heteroscedastic_gp`	kernel	rbf set by this job, not by the shared spec
	likelihood_noise	0.001 starting value; it is learned
	outputscale	1.0
	apply_outputscale	True
	max_train_n	5000 molecules the process is fitted on; an exact Gaussian process is cubic in this. The test set is untouched and the number actually fitted is written into every results row
	fallback_adam_lr	0.1 used only when the standard fitter refuses the model
	fallback_adam_iters	100 used only when the standard fitter refuses the model
	single_thread_fit	False a threading workaround, off since the environment check found one runtime
	init_lengthscale_from_data	True the RBF length scale starts at the median distance between training molecules, not at the library default
	lengthscale_probe_n	500 molecules sampled to estimate that median

Configuration	Hyperparameter	Default value
	collapse_fraction	0.05 a fit whose predictions vary by less than this fraction of the training label spread is recorded as collapsed rather than scored
	noise network hidden sizes	[64, 64] models/models.py, not in the shared spec
	noise network output	Softplus, plus 10^{-4} models/models.py, not in the shared spec
	Adam learning rate, process	0.1 models/models.py, not in the shared spec
	Adam learning rate, noise network	0.001 models/models.py, not in the shared spec
	noise network epochs	100 the shared epoch cap
Neural networks		
NN-α `dnn`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
NN-β `mlp`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
BNN-α `dnn_bnn_full`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo 1 / number of training molecules
BNN-β `mlp_bnn_full`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1

Configuration	Hyperparameter	Default value
	bnn_kl_weight	elbo 1 / number of training molecules
VBLL-α `dnn_bnn_full_variational`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
VBLL-β `mlp_bnn_full_variational`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
VBLL-α (noise head) `dnn_bnn_full_variational_hetero`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0
	observation noise	predicted per molecule a variational noise head on the same features, started at the single-number value above
VBLL-β (noise head) `mlp_bnn_full_variational_hetero`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	vbll_prior_mu	0.0
	vbll_prior_sigma	1.0
	vbll_init_log_sigma	-3.0
	vbll_init_log_noise_var	-2.0

Configuration	Hyperparameter	Default value
	observation noise	predicted per molecule a variational noise head on the same features, started at the single-number value above
BNN-α (variance head) `dnn_bnn_full_mve`	hidden_sizes	[128, 64]
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo 1 / number of training molecules
	output head	value and log variance two outputs
	loss	Gaussian negative log likelihood
	log variance clamp	$[-10, 10]$ scripts/loss_functions.py, not in the shared spec
BNN-β (variance head) `mlp_bnn_full_mve`	hidden_size	128
	num_hidden_layers	2
	dropout_rate	0.2
	activation	relu
	bnn_prior_mu	0.0
	bnn_prior_sigma	0.1
	bnn_kl_weight	elbo 1 / number of training molecules
	output head	value and log variance two outputs
	loss	Gaussian negative log likelihood
	log variance clamp	$[-10, 10]$ scripts/loss_functions.py, not in the shared spec

Table S.2: Table B: the tuned settings that replace a default, and the four datasets they apply to. A setting is chosen once per model per dataset, not per representation, and is then used for all six representations: a two-way comparison across model and representation cannot be read if each cell carries its own hyperparameters. Only the Bayesian and variational networks are tuned; every other configuration in Table A trains at the default. A model with no row for a dataset kept the default on that dataset. Learning rates are printed to three significant figures; the source files hold the full value. Source files: `results/master_tuned_hyperparameters.json` for QM9 and `results/master_tuned_hyperparameters_lab.json` for the three assay datasets.

Dataset	Configuration	Hyperparameter	Tuned value
QM9	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
QM9	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
QM9	VBLL-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
QM9	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
QM9	BNN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
QM9	VBLL-β	dropout_rate	0.357
		hidden_size	64
		lr	0.00119
		num_hidden_layers	1
Caco-2	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
Caco-2	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
Caco-2	VBLL-α	activation	relu
		hidden_size1	64
		hidden_size2	32
Caco-2	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
Caco-2	BNN-β	dropout_rate	0.401
		hidden_size	64

Dataset	Configuration	Hyperparameter	Tuned value
		lr	0.00587
		num_hidden_layers	3
hERG K_i	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
hERG K_i	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
hERG K_i	VBLL-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
hERG K_i	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
hERG K_i	BNN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
LogD	NN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
LogD	BNN-α	activation	tanh
		hidden_size1	64
		hidden_size2	64
LogD	VBLL-α	activation	tanh
		hidden_size1	256
		hidden_size2	32
LogD	NN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
LogD	BNN-β	dropout_rate	0.401
		hidden_size	64
		lr	0.00587
		num_hidden_layers	3
LogD	VBLL-β	dropout_rate	0.079
		hidden_size	128
		lr	0.00011
		num_hidden_layers	2

Table S.3: Table C: the 200 descriptors that make up the PDV representation. One entry is one feature of the vector, in the order the pipeline computes them. The list is fixed in the code rather than taken from RDKit's current default set, which grows between releases, so the same 200 features are built on every dataset and in every run. They were computed with RDKit 2022.09.5. Descriptors that return a non-finite value on a molecule have that value replaced by zero before the standardization constants are computed. Source: `DEFAULT_DESCRIPTOR_LIST` in `scripts/process_and_train.py`.

BalabanJ	NOCount	SlogP_VSA7	fr_bicyclic
BertzCT	NumAliphaticCarbocycles	SlogP_VSA8	fr_diazo
Chi0	NumAliphaticHeterocycles	SlogP_VSA9	fr_dihydropyridine
Chi0n	NumAliphaticRings	TPSA	fr_epoxide
Chi0v	NumAromaticCarbocycles	VSA_EState1	fr_ester
Chi1	NumAromaticHeterocycles	VSA_EState10	fr_ether
Chi1n	NumAromaticRings	VSA_EState2	fr_furan
Chi1v	NumHAcceptors	VSA_EState3	fr_guanido
Chi2n	NumHDonors	VSA_EState4	fr_halogen
Chi2v	NumHeteroatoms	VSA_EState5	fr_hdrzine
Chi3n	NumRadicalElectrons	VSA_EState6	fr_hdrzone
Chi3v	NumRotatableBonds	VSA_EState7	fr_imidazole
Chi4n	NumSaturatedCarbocycles	VSA_EState8	fr_imide
Chi4v	NumSaturatedHeterocycles	VSA_EState9	fr_isocyan
EState_VSA1	NumSaturatedRings	fr_Al_COO	fr_isothiocyan
EState_VSA10	NumValenceElectrons	fr_Al_OH	fr_ketone
EState_VSA11	PEOE_VSA1	fr_Al_OH_noTert	fr_ketone_Topliss
EState_VSA2	PEOE_VSA10	fr_ArN	fr_lactam
EState_VSA3	PEOE_VSA11	fr_Ar_COO	fr_lactone
EState_VSA4	PEOE_VSA12	fr_Ar_N	fr_methoxy
EState_VSA5	PEOE_VSA13	fr_Ar_NH	fr_morpholine
EState_VSA6	PEOE_VSA14	fr_Ar_OH	fr_nitrile
EState_VSA7	PEOE_VSA2	fr_COO	fr_nitro
EState_VSA8	PEOE_VSA3	fr_COO2	fr_nitro_arom
EState_VSA9	PEOE_VSA4	fr_C_O	fr_nitro_arom_nonortho
ExactMolWt	PEOE_VSA5	fr_C_O_noCOO	fr_nitroso
FpDensityMorgan1	PEOE_VSA6	fr_C_S	fr_oxazole
FpDensityMorgan2	PEOE_VSA7	fr_HOCCN	fr_oxime
FpDensityMorgan3	PEOE_VSA8	fr_Imine	fr_para_hydroxylation
FractionCSP3	PEOE_VSA9	fr_NH0	fr_phenol
HallKierAlpha	RingCount	fr_NH1	fr_phenol_noOrthoHbond
HeavyAtomCount	SMR_VSA1	fr_NH2	fr_phos_acid
HeavyAtomMolWt	SMR_VSA10	fr_N_O	fr_phos_ester
Ipc	SMR_VSA2	fr_Ndealkylation1	fr_piperdine
Kappa1	SMR_VSA3	fr_Ndealkylation2	fr_piperzine
Kappa2	SMR_VSA4	fr_Nhpyrrole	fr_priamide
Kappa3	SMR_VSA5	fr_SH	fr_prisulfonamd
LabuteASA	SMR_VSA6	fr_aldehyde	fr_pyridine
MaxAbsEStateIndex	SMR_VSA7	fr_alkyl_carbamate	fr_quatN
MaxAbsPartialCharge	SMR_VSA8	fr_alkyl_halide	fr_sulfide
MaxEStateIndex	SMR_VSA9	fr_allylic_oxid	fr_sulfonamd
MaxPartialCharge	SlogP_VSA1	fr_amide	fr_sulfone
MinAbsEStateIndex	SlogP_VSA10	fr_amidine	fr_term_acetylene

MinAbsPartialCharge	SlogP_VSA11	fr_aniline	fr_tetrazole
MinEStateIndex	SlogP_VSA12	fr_aryl_methyl	fr_thiazole
MinPartialCharge	SlogP_VSA2	fr_azide	fr_thiocyan
MolLogP	SlogP_VSA3	fr_azo	fr_thiophene
MolMR	SlogP_VSA4	fr_barbitur	fr_unbrch_alkane
MolWt	SlogP_VSA5	fr_benzene	fr_urea
NHOHCount	SlogP_VSA6	fr_benzodiazepine	qed

Additional file 2: Configurations excluded from the robustness analysis

For each model and representation on the QM9 HOMO–LUMO gap, the number of noise strategies for which the configuration was excluded because its clean-label R^2 was too low.

Table S.4: Configurations excluded from robustness analysis due to baseline $R^2 \leq 0.6$ on the QM9 HOMO–LUMO gap, summarized by model and representation. Each cell shows the number of noise strategies (out of 6) for which the configuration was excluded.

Model	Mol2Vec	MHGGNN	SMILES	Rand. SMILES
BNN-α	6	6	—	—
BNN-β	6	6	—	—
VBLL-α	6	6	—	—
VBLL-β	6	6	—	—
RF	—	—	—	4
QRF	—	—	—	4
NGBoost	—	—	6	4

Additional file 3: Spearman correlations between the robustness profiles of the models

Pairs of models whose noise degradation slope profiles on the QM9 HOMO–LUMO gap correlate at $\rho \geq 0.95$, and which model of each pair was left out of the ANOVA.

Table S.5: Pairwise Spearman rank correlations between model noise degradation slope (NDS) profiles on the QM9 HOMO–LUMO gap ($\rho \geq 0.95$). "Model A" and "Model B" label each pair, both treated symmetrically. Checkmarks indicate pairs where one model was excluded from the ANOVA due to near-redundancy. Gauche (RBF) pairs omitted (all $\rho = 1.0$, $n = 6$, PDV-only artifact).

Model A	Model B	n	ρ	Excluded
VBLL-β	NGBoost	18	0.998	
LightGBM	XGBoost	46	0.998	
Gauche	VBLL-β	18	0.996	✓
QRF	RF	40	0.995	✓

Model A	**Model B**	n	ρ	**Excluded**
VBLL-α	LightGBM	24	0.992	
RF	XGBoost	46	0.989	
NGBoost	XGBoost	40	0.988	
NGBoost	RF	40	0.987	
LightGBM	RF	46	0.986	
VBLL-β	SVM	24	0.986	
QRF	XGBoost	40	0.986	✓
NGBoost	QRF	34	0.986	✓
VBLL-α	NGBoost	18	0.986	
VBLL-α	XGBoost	24	0.984	
LightGBM	NGBoost	40	0.984	
LightGBM	QRF	40	0.984	✓
VBLL-α	VBLL-β	24	0.983	
NN-α	NN-β	52	0.983	
LightGBM	SVM	46	0.982	
BNN-β	SVM	34	0.981	
VBLL-α	SVM	24	0.979	
BNN-α	VBLL-β	24	0.979	
VBLL-α	QRF	24	0.978	✓
SVM	XGBoost	46	0.978	
BNN-α	BNN-β	36	0.977	
BNN-α	SVM	34	0.976	
VBLL-α	Gauche	18	0.975	✓
VBLL-α	RF	24	0.975	
QRF	SVM	40	0.974	✓
BNN-α	VBLL-α	24	0.974	
LightGBM	VBLL-β	24	0.973	
VBLL-α	BNN-β	24	0.972	
Gauche	SVM	40	0.972	✓
Gauche	XGBoost	44	0.970	✓
NGBoost	SVM	40	0.970	
NN-α	BNN-β	36	0.970	
Gauche	LightGBM	42	0.969	✓
RF	SVM	46	0.969	
BNN-β	VBLL-β	24	0.968	
VBLL-β	QRF	24	0.966	✓
NN-α	BNN-α	36	0.965	
VBLL-β	XGBoost	24	0.965	
NN-β	SVM	46	0.962	
NN-α	SVM	46	0.960	
Gauche	NGBoost	34	0.959	✓
LightGBM	BNN-β	36	0.959	
BNN-α	NN-β	36	0.958	
BNN-α	QRF	28	0.956	✓
Gauche	QRF	34	0.956	✓
VBLL-β	RF	24	0.955	
Gauche	RF	40	0.954	✓
BNN-β	XGBoost	34	0.953	
BNN-β	QRF	28	0.952	✓
NN-α	VBLL-β	24	0.952	

Model A	Model B	n	ρ	Excluded
NN-β	BNN-β	36	0.952	
BNN-α	LightGBM	36	0.951	
NN-β	VBLL-β	24	0.950	
BNN-α	Gauche	30	0.950	✓

Additional file 4: Spearman correlations between the robustness profiles of the representations

Pairwise correlations between the noise degradation slope profiles of the representations on the QM9 HOMO–LUMO gap, and which representation of each pair was left out of the ANOVA.

Table S.6: Pairwise Spearman rank correlations between representation noise degradation slope (NDS) profiles on the QM9 HOMO–LUMO gap. "Rep A" and "Rep B" label each pair of molecular representations, both treated symmetrically. Checkmarks indicate pairs where one representation was excluded from the ANOVA.

Rep. A	Rep. B	n	ρ	Excluded
ECFP4	Morgan	60	0.995	✓
MHGGNN	Morgan	48	0.994	✓
Rand. SMILES	SMILES	16	0.994	✓
Rand. SMILES	SNS	16	0.994	✓
ECFP4	MHGGNN	54	0.993	
ECFP4	SNS	56	0.992	✓
Morgan	SNS	52	0.992	✓
MHGGNN	SNS	44	0.991	✓
PDV	ECFP4	72	0.990	
PDV	MHGGNN	48	0.989	
SMILES	SNS	52	0.988	✓
Morgan	Rand. SMILES	16	0.988	✓
Morgan	SMILES	54	0.988	✓
PDV	Morgan	54	0.986	✓
PDV	Rand. SMILES	12	0.986	✓
PDV	SNS	50	0.984	✓
ECFP4	SMILES	72	0.984	
Mol2Vec	Morgan	48	0.982	✓
MHGGNN	Mol2Vec	54	0.982	
ECFP4	Mol2Vec	54	0.982	
PDV	PDV (binary)	72	0.980	✓
MHGGNN	SMILES	48	0.980	
PDV	Mol2Vec	48	0.979	
ECFP4	Rand. SMILES	16	0.979	✓
PDV	SMILES	66	0.978	
MHGGNN	PDV (binary)	54	0.972	✓
Mol2Vec	SNS	44	0.969	✓
Mol2Vec	SMILES	48	0.966	
ECFP4	PDV (binary)	78	0.965	✓

Rep. A	Rep. B	n	ρ	Excluded
Morgan	PDV (binary)	60	0.962	✓
PDV (binary)	SNS	56	0.961	✓
Mol2Vec	PDV (binary)	54	0.956	✓
MHGGNN	Rand. SMILES	16	0.956	✓
PDV (binary)	SMILES	72	0.941	✓
PDV (binary)	Rand. SMILES	16	0.929	✓
Mol2Vec	Rand. SMILES	16	0.891	✓

Additional file 5: Intraclass correlation coefficients for model pairs

ICC(1,1) of the noise degradation slope profiles for the model pairs with ICC ≥ 0.5, on the QM9 HOMO–LUMO gap.

Table S.7: Intraclass correlation coefficients (ICC(1,1)) for model pairs with ICC ≥ 0.5, assessing within-family consistency of noise degradation slope (NDS) profiles on the QM9 HOMO–LUMO gap. ICC(1,1) measures the proportion of total variance attributable to between-configuration differences; values near 1.0 indicate that two models rank configurations almost identically. Lower (more negative) NDS indicates faster degradation under noise. The Family column groups each pair: "Tree" (both tree-based), "α/β" (architecture-pair partners such as BNN-α/BNN-β), "α" (both share the α neural-network backbone), or "Cross" (cross-family pair).

Model A	Model B	n	ICC(1,1)	Mean $\lvert\Delta$NDS$\rvert$	Family
LightGBM	XGBoost	8	0.944	0.0060	Tree
BNN-α	BNN-β	6	0.846	0.0101	α/β
QRF	RF	7	0.810	0.0216	Tree
VBLL-α	VBLL-β	4	0.782	0.0108	α/β
QRF	XGBoost	7	0.754	0.0157	Tree
LightGBM	QRF	7	0.743	0.0133	Tree
VBLL-α	LightGBM	4	0.698	0.0115	Cross
VBLL-β	SVM	4	0.689	0.0133	Cross
RF	XGBoost	8	0.678	0.0212	Tree
VBLL-α	XGBoost	4	0.656	0.0133	Cross
BNN-α	VBLL-α	4	0.604	0.0134	α
LightGBM	RF	8	0.586	0.0244	Tree
LightGBM	VBLL-β	4	0.577	0.0174	Cross
LightGBM	SVM	8	0.561	0.0117	Cross
BNN-α	VBLL-β	4	0.552	0.0123	α/β
VBLL-β	XGBoost	4	0.544	0.0195	Cross
VBLL-α	QRF	4	0.540	0.0229	Cross
VBLL-α	SVM	4	0.522	0.0124	Cross
Gauche	VBLL-β	3	0.519	0.0113	Cross

Additional file 6: Global overview of noise robustness on ECFP4

Noise robustness of every model on the ECFP4 representation, on the QM9 HOMO–LUMO gap.

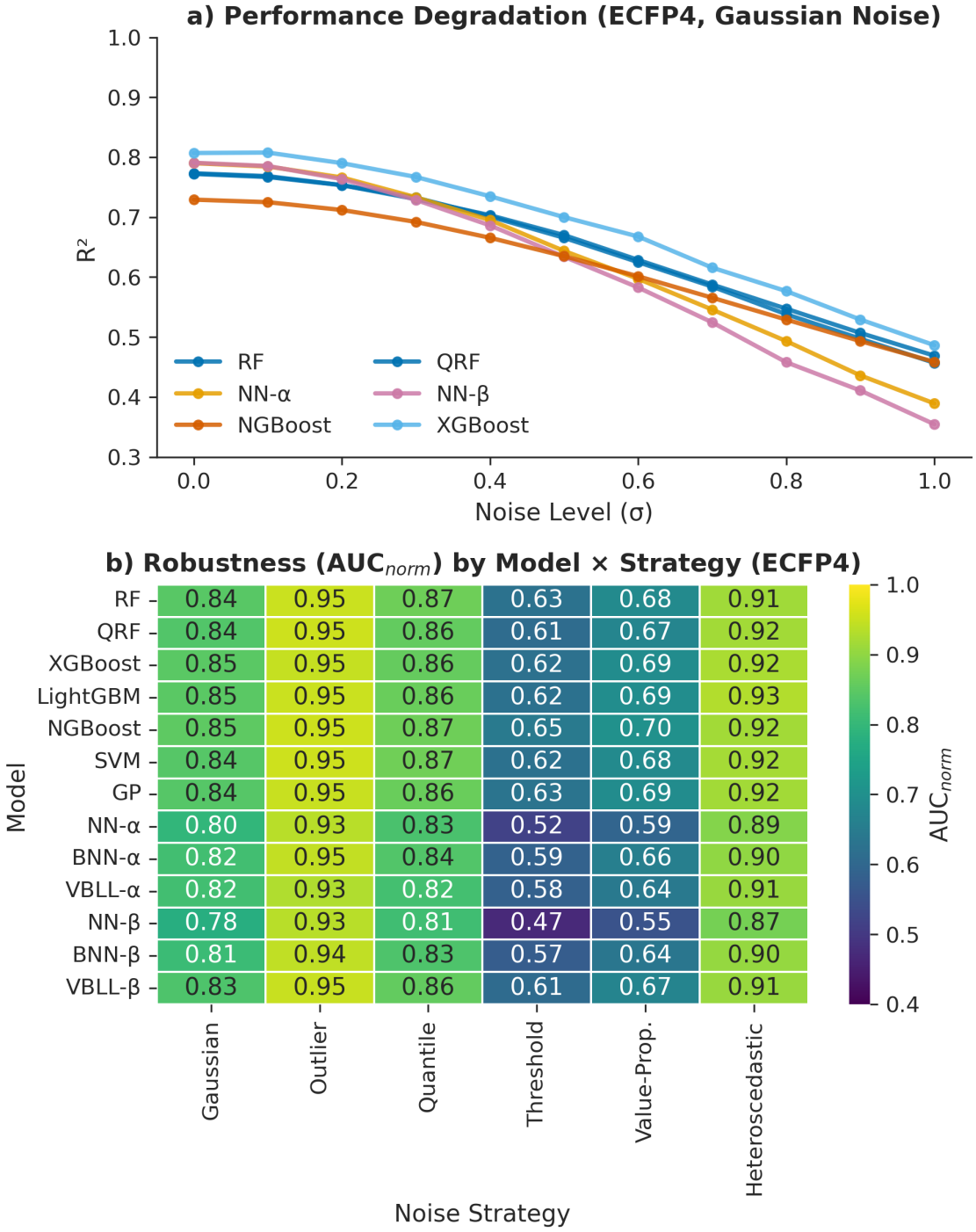


Figure S.1: Global overview of noise robustness on the ECFP4 representation, evaluated on QM9 HOMO–LUMO gap. a) Performance degradation curves (R^2 versus noise level σ) under Gaussian noise. b) Noise degradation slope (NDS) heatmap across model architectures and noise strategies; lower (more negative) NDS indicates faster degradation under noise.

Additional file 7: Clean R^2 against robustness on PDV under Gaussian noise

Clean-label R^2 against the noise degradation slope for every model on PDV under

Gaussian noise, on the QM9 HOMO–LUMO gap.

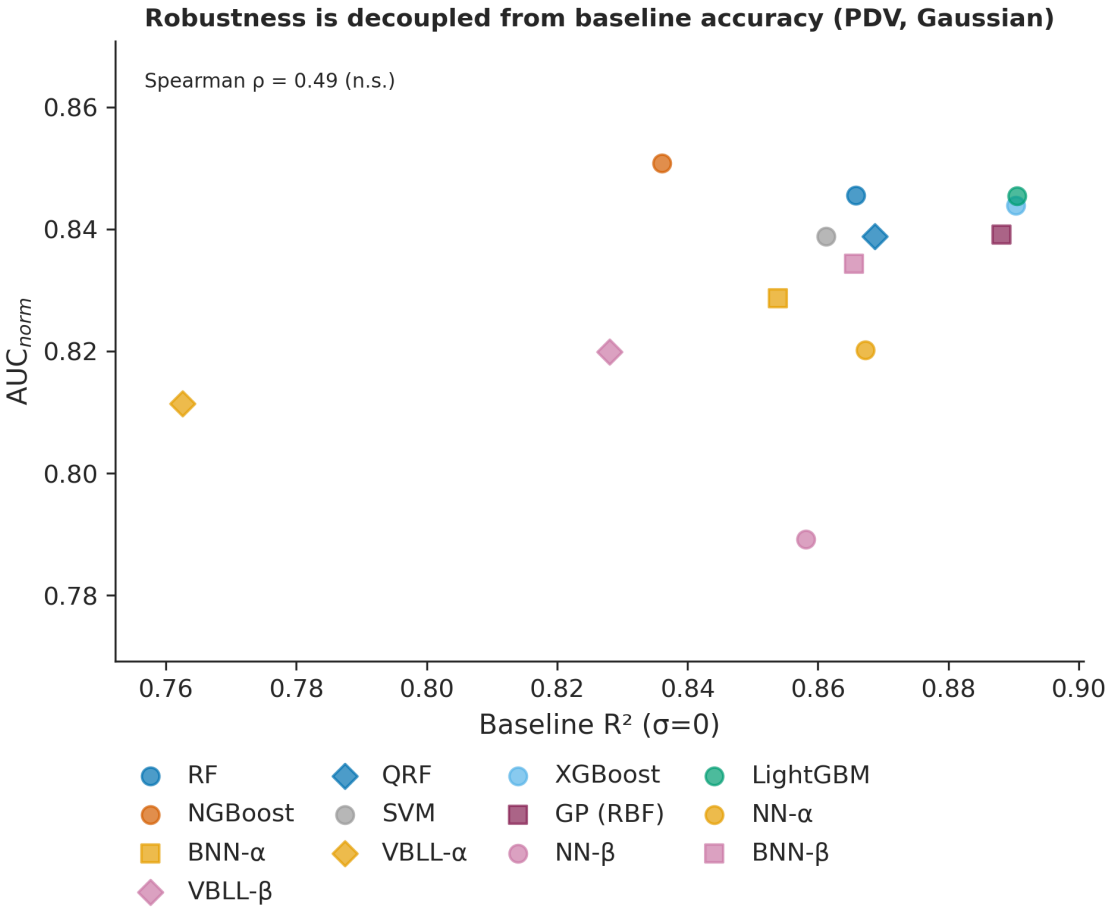


Figure S.2: Baseline R^2 versus noise degradation slope (NDS) scatter for all models on the physicochemical descriptor vector (PDV) representation under Gaussian noise, on the QM9 HOMO–LUMO gap. Lower (more negative) NDS indicates faster performance loss under noise.

Additional file 8: Bayesian transformation effects by representation

The effect of each Bayesian transformation of the neural networks on the noise degradation slope, by representation, on the QM9 HOMO–LUMO gap.

Table S.8: Neural network Bayesian transformation effects on noise degradation slope (NDS) by representation, on the QM9 HOMO–LUMO gap. Positive Δ NDS indicates the Bayesian variant is more robust (less negative slope) than its deterministic counterpart.

Model	Representation	Baseline R^2	NN-β NDS	BNN-β NDS	Δ NDS
NN-β	ECFP4	0.791	−0.464	−0.388	+0.076
NN-β	PDV	0.858	−0.487	−0.382	+0.106
NN-β	SMILES	0.721	−0.558	−0.365	+0.193

Additional file 9: Uncertainty metrics for probabilistic models on ECFP4

Uncertainty metrics of the probabilistic models on ECFP4 under Gaussian noise, on the QM9 HOMO–LUMO gap.

Table S.9: Uncertainty quantification metrics for probabilistic models on the ECFP4 representation (Gaussian noise strategy) on the QM9 HOMO–LUMO gap. Unc-Error ρ: Spearman correlation between predicted uncertainty and absolute prediction error. Unc-Noise ρ: Spearman correlation between predicted uncertainty and injected noise level σ. ECE: expected calibration error (lower = better-calibrated). Coverage targets: 68% (1σ), 95% (2σ).

Model	Unc-Error ρ	Unc-Noise ρ	ECE	Cov. 1σ	Cov. 2σ
QRF	**0.25**	0.27	0.19	74.7%	94.6%
NGBoost	0.22	**0.37**	**0.11**	69.2%	94.2%
GP	0.18	0.34	0.23	78.6%	96.0%
BNN-β	0.20	0.37	0.29	78.6%	95.2%
BNN-α	0.15	0.32	0.24	76.6%	95.1%
VBLL-β	0.19	0.34	0.19	73.9%	95.5%
VBLL-α	0.11	0.31	0.22	74.5%	95.1%

Additional file 10: Variance decomposition on the three assay datasets

ANOVA variance decomposition (η^2, in percent) on LogD, Caco-2 efflux and hERG K_i.

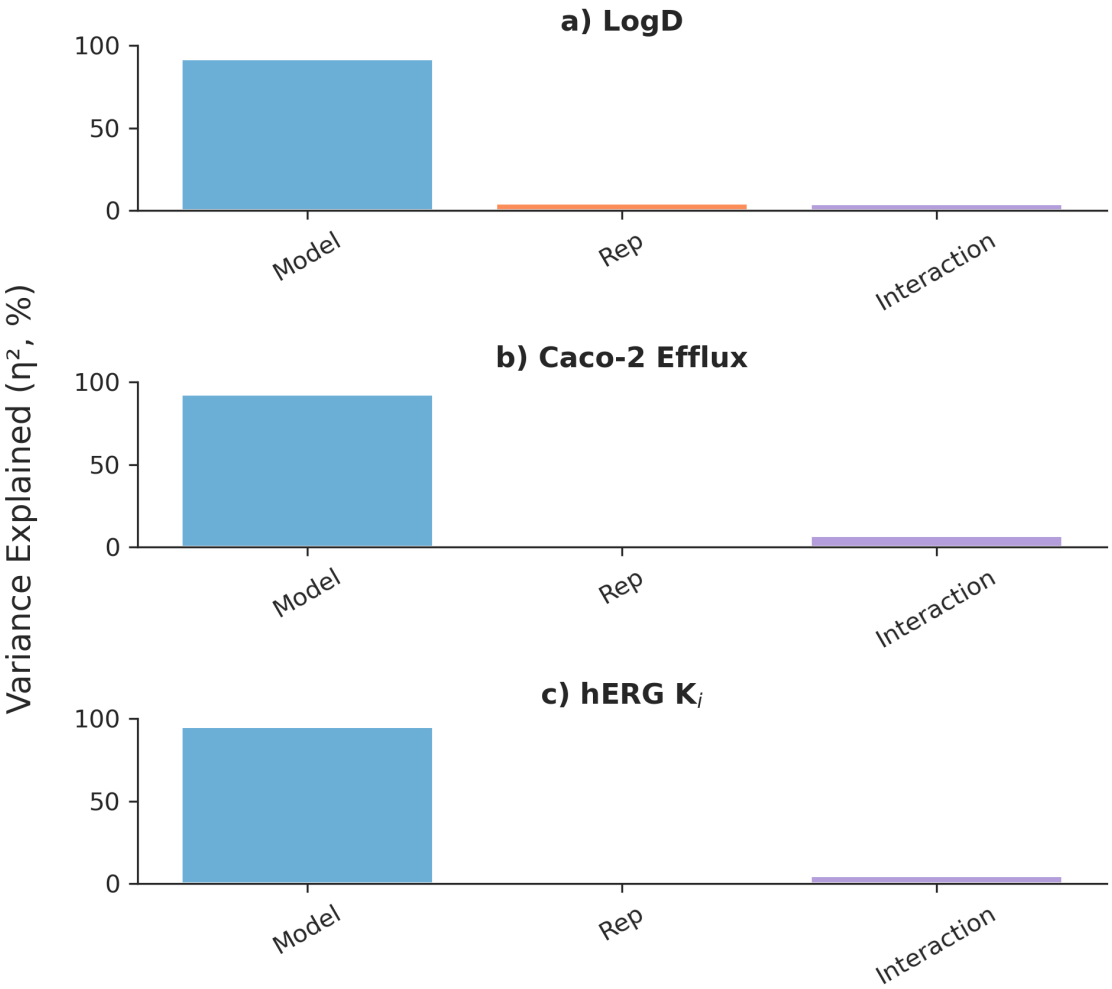


Figure S.3: ANOVA variance decomposition (η^2, %) on external validation datasets: a) LogD, b) Caco-2 Efflux, and c) hERG K$_i$. Model, representation, and interaction contributions to noise degradation slope (NDS) variance for each dataset; lower (more negative) NDS indicates faster performance loss under noise.

Additional file 11: Random forest against quantile regression forest on the three assay datasets

The noise degradation slope of random forest and of quantile regression forest on LogD, Caco-2 efflux and hERG K_i.

Table S.10: Random forest (RF) vs quantile regression forest (QRF) noise degradation slope (NDS) on the three external validation datasets (LogD, Caco-2 Efflux, hERG K$_i$). Lower (more negative) NDS indicates faster degradation under noise.

Dataset	RF NDS	QRF NDS	Δ NDS
ChEMBL-hERG-Ki	−0.231	−0.359	−0.128
OpenADMET-Caco2	−0.534	−0.627	−0.094
OpenADMET-LogD	−0.101	−0.185	−0.083

S.2 The flaws of the similarity assumption

One row of Table S.11 is one representation on one dataset. Representations are listed in each dataset's pool order.

Table S.11: All-pairs correlation, five-nearest-neighbor prediction and activity cliffs for every representation in each regression dataset's pool. All pairs: Spearman correlation between pair distance and absolute label difference, on a fixed sample of 2,000 molecules (seed 0) where a dataset has more. Five neighbors: Spearman correlation and RMSE (in the label's units) of the mean label of the five nearest molecules from other scaffold groups. Cliff molecules: molecules in at least one of this representation's closest pairs whose labels differ by more than the dataset's gap (Table 4.4). The number of closest pairs is the number ECFP4 has at Tanimoto ≥ 0.55, a few more where distances tie at the cut-off. Shared: percentage of ECFP4's similar pairs that are also among this representation's closest pairs. HLM clearance's pool has no ECFP4.

Dataset	Representation	All pairs Spearman	Five neighbors Spearman	Five neighbors RMSE	Cliff molecules	Shared (%)
hERG K_i	MHG-GNN	0.07	0.71	0.65	208	67
	Mordred	0.07	0.72	0.62	165	62
	WL graph kernel	0.04	0.71	0.64	267	74
	Avalon	0.02	0.69	0.65	238	63
	GROVER	0.05	0.70	0.63	192	63
	PDV	0.08	0.73	0.62	208	61
	ECFP4	0.06	0.72	0.63	211	100
	Atom pairs	0.04	0.74	0.60	138	68
	ChemBERT	0.07	0.68	0.66	166	57
	SMI-TED	0.09	0.69	0.65	187	58
LogD	MHG-GNN	0.09	0.81	0.66	286	42
	Mordred	0.06	0.80	0.68	308	35
	GROVER	0.09	0.81	0.67	399	51
	ECFP4	0.12	0.76	0.74	731	100
	SMI-TED	0.10	0.76	0.74	758	42
	MoLFormer	0.13	0.77	0.73	768	46
	Avalon	0.04	0.73	0.77	801	49
	Topological torsions	0.10	0.74	0.76	808	71
	ChemBERT	0.09	0.75	0.75	808	42
	GraphMVP	0.11	0.76	0.74	415	38
Caco-2 efflux	WL graph kernel	0.13	0.58	0.35	398	54
	MHG-GNN	0.16	0.60	0.34	347	53
	ECFP4	0.08	0.59	0.34	324	100
	PDV	0.18	0.62	0.34	237	43
	GROVER	0.06	0.61	0.34	302	55
	Avalon	−0.00	0.56	0.35	335	46
	Topological torsions	0.15	0.61	0.34	348	74
	SMI-TED	0.14	0.57	0.35	400	44
	Atom pairs	0.08	0.59	0.34	315	57
	MoLFormer	0.09	0.59	0.34	311	51
RLM clearance	PDV	0.06	0.56	0.62	433	34

Dataset	Representation	All pairs Spearman	Five neighbors Spearman	Five neighbors RMSE	Cliff molecules	Shared (%)
	MHG-GNN	0.02	0.47	0.67	487	46
	SMI-TED	0.04	0.46	0.67	483	48
	Mordred	0.07	0.49	0.66	445	42
	MoLFormer	0.08	0.44	0.68	412	60
	GROVER	0.05	0.43	0.69	436	46
	WL graph kernel	0.00	0.40	0.71	554	36
	Atom pairs	0.07	0.47	0.67	437	60
	Avalon	0.04	0.29	0.74	389	55
	ECFP4	0.02	0.36	0.71	458	100
MDR1–MDCK efflux	Mordred	0.16	0.58	0.57	363	45
	PDV	0.23	0.64	0.54	302	39
	MHG-GNN	0.20	0.52	0.62	344	48
	Avalon	−0.04	0.50	0.61	378	61
	GROVER	−0.03	0.47	0.63	356	49
	SMI-TED	0.10	0.59	0.56	389	51
	Atom pairs	0.00	0.58	0.57	374	64
	WL graph kernel	0.17	0.47	0.64	413	40
	MACCS keys	−0.04	0.54	0.59	379	46
	ECFP4	0.02	0.51	0.61	397	100
HLM clearance	Mordred	0.07	0.49	0.55	429	43
	PDV	0.10	0.55	0.52	390	34
	MHG-GNN	0.08	0.46	0.56	435	46
	SMI-TED	0.03	0.45	0.56	475	48
	MoLFormer	0.00	0.43	0.56	422	60
	GROVER	−0.03	0.41	0.59	413	46
	Atom pairs	−0.04	0.47	0.55	426	60
	ChemBERT	−0.02	0.39	0.58	452	45
	WL graph kernel	0.12	0.38	0.59	490	37
	Avalon	0.01	0.32	0.61	416	56
QM9	Mordred	0.18	0.84	0.68	1,443	3
	Avalon	0.19	0.83	0.70	1,148	9
	SMI-TED	0.13	0.85	0.65	1,576	13
	ECFP4	0.19	0.88	0.64	1,495	100
	SELFormer	0.20	0.82	0.73	300	2
	ChemBERTa-2	0.19	0.89	0.61	1,108	1
	Mol2vec	0.14	0.86	0.66	1,274	24
	ChemBERT	0.19	0.83	0.72	1,295	7
	GraphMVP	0.20	0.83	0.71	1,231	9
	MoLFormer	0.21	0.84	0.70	1,423	3

S.3 Building adaptive hybrid representations

Additional file 12: Settings of every model, as downstream model and as internal evaluator

One row is one model. The middle two columns give its settings when it scores a hybrid (downstream model) and when it scores one representation inside the search (internal evaluator). The models fitted only to compute feature importance are described in Chapter 5.

Table S.12: Settings of every model. p is the number of features. "Molecules" is the number of training molecules the internal evaluator is fitted on; above it, a random subset of that size is drawn with a fixed seed. Settings not listed are the library's defaults. Every setting is the same on every dataset. The Gaussian process runs only on the datasets named in Chapter 5, "Models and metrics".

Model	As downstream model	As internal evaluator	Molecules	Library; source
Random forest	100 trees, no depth limit; $\sqrt{p}$ features tried at each split	50 trees, depth at most 10; all p features tried at each split on regression, $\sqrt{p}$ on classification	all	scikit-learn [1]; Breiman [2]
LightGBM	100 trees	50 trees, depth at most 4	all	lightgbm; Ke et al. [3]
XGBoost	100 trees; half of the features sampled for each tree	50 trees, depth at most 4	all	xgboost; Chen and Guestrin [4]
SVM	RBF kernel, $C = 1$. Not fitted above 10,000 molecules, or above 5,000 under uncapped augmentation	RBF kernel, $C = 1$	at most 500	scikit-learn; Vapnik [5]
Gaussian process	Regression: RBF plus white-noise kernel, labels normalized, 3 optimizer restarts. Classification: RBF kernel. Not fitted above 1,000 molecules	Regression: scaled RBF kernel, fixed noise variance 0.01, no restarts. Classification: scaled RBF kernel, no restarts	at most 500	scikit-learn; Rasmussen and Williams [6]

Model	As downstream model	As internal evaluator	Molecules	Library; source
DNN	Hidden layers of 256, 128 and 64 units, ReLU, dropout 0.2 [7]; Adam [8] at learning rate 0.001, halved after 5 epochs without improvement; batch size 256; up to 200 epochs; stops after 20 epochs without improvement on 20% of the training molecules that share no group with the rest, using the outer split's grouping; labels z-normalized	The same, except that the early-stopping 20% is drawn at random	all	PyTorch [9]

Additional file 13: Downstream models left out, by dataset and augmentation setting

One row is one rule that leaves a downstream model out. Only the 16 datasets of Table 5.8 are listed; a dataset and setting not named in any row runs every downstream model in Table 5.6.

Table S.13: Downstream models left out, by dataset and augmentation setting. Every rule was set so that a search finishes within the time limit of one job; none was set from scores. hERG K_i drops no model under any setting.

Model left out	Augmentation setting	Datasets
XGBoost	all three	H3K4me3, TATA promoter
SVM	all three	datasets above 10,000 molecules: H3K4me3
SVM	uncapped	datasets above 5,000 molecules: QM9, LogD[a], TATA promoter[a]
SVM	uncapped	Koenig expression
SVM	capped	Caco-2 efflux, LogD, rat liver microsome clearance, MDR1–MDCK efflux, Buchwald–Hartwig plates
Random forest and SVM	capped	QM9
Random forest	uncapped	QM9, LogD
Gaussian process	all three	every dataset except humanness (Chapter 5, "Models and metrics")

[a]Counted on the whole dataset: LogD has 5,028 molecules and TATA promoter 5,274. A

search run inside one outer fold is given about 80% of them and keeps the SVM.

Additional file 14: Size of the search

One row is one downstream model. The counts are per dataset, objective and augmentation setting, with adaptive pruning off.

Table S.14: Number of configurations searched per dataset, objective and augmentation setting. A configuration is one choice of downstream model, internal evaluator, allocation method, importance method, representation count and budget. At the full tier each (downstream model, internal evaluator) pair runs 5 allocation methods × 3 importance methods × 4 representation counts × 8 budgets = 480 configurations; at the fast tier it runs $3 \times 1 \times 2 \times 8 = 48$ (Table 5.6). "Matched" is counted once where it repeats an evaluator already in the list. Each configuration is scored on 5 folds, so the full tier fits a downstream model 40,800 times (50,400 on humanness) and the fast tier 4,080 times. Where a dataset leaves a downstream model out (Additional file 13), its pairs drop out: a fast search on H3K4me3 has 9 pairs, so 432 configurations.

Downstream model	Internal evaluators	Full tier	Fast tier
Random forest	3 (matched is random forest)	1,440	144
LightGBM	3 (matched is LightGBM)	1,440	144
DNN	3 (matched is DNN)	1,440	144
XGBoost	4	1,920	192
SVM	4	1,920	192
Gaussian process (humanness only)	4	1,920	not run
Total	17 pairs (21 on humanness)	8,160 (10,080 on humanness)	816

Additional file 15: Design experiments behind the search settings

One row is one setting of the search, or one rule that was tested and then fixed instead of searched. Every number comes from preliminary design runs made before the main experiments; they are not results of the thesis, and several used other budgets than the final search.

Table S.15: Design experiments behind the search settings. Preliminary design runs, not the thesis's full runs.

Setting	Values compared	Datasets, folds, budgets	What was measured, and the result	Values kept (full / fast)
Downstream model	Nine models, among them random forest, LightGBM, XGBoost, CatBoost, DNN, k-nearest neighbors and a Gaussian process, each on each of 10 base representations at full width	Buchwald–Hartwig yield, Caco-2 efflux, Koenig expression, LogD, breast cancer PAM50 subtype; 5 folds	Best held-out score over the 10 representations, and seconds per fit. The DNN scored best on 4 of the 5 datasets and third on Caco-2 (0.2936 against 0.2851 RMSE, $\log_{10}$ efflux ratio)	Full: random forest, LightGBM, XGBoost, SVM, DNN, and the Gaussian process on humanness / fast: the same five
Budget	None compared	–	–	10, 25, 50, 100, 250, 500, 1,000 and 2,000 features at both tiers
Representation count	2, 3, 5 and all 10	9 datasets, 3 tree-ensemble downstream models, budgets 500 and 1,000 features; 5 folds	The best count for each (downstream model, budget) combination	Full: 2, 3, 5, all / fast: 3, all
Allocation method	8 candidates: the five searched and 3 further pooled candidates, each at every representation count	The same 9 datasets, models and budgets: 54 (dataset, model, budget) combinations	Each candidate was scored at its best representation count, and the number of combinations in which it was in the top 3 of the 8 was counted. Fixed shares: linear 30, squared 27, rank decay 26, softmax 25; global pool with squared weights 14	Full: the five described in Chapter 5 / fast: linear, squared, rank decay

Setting	Values compared	Datasets, folds, budgets	What was measured, and the result	Values kept (full / fast)
Importance method, agreement	8 methods	LogD, hERG K_i, Buchwald–Hartwig yield, nanobody thermostability, antibody FcRn retention, breast cancer PAM50 subtype, lung adenocarcinoma vs squamous; each method fitted once on the whole dataset	Spearman correlation between two methods' scores over every feature. Kept pairs: LightGBM gain and TreeSHAP 0.20–0.64, LightGBM gain and DeepLIFT 0.21–0.67, TreeSHAP and DeepLIFT 0.56–0.99 across the 7 datasets	One method kept from each group whose rankings agree above 0.95
Importance method, score	28 methods, among them three controls: feature variance, Laplacian score and random ranking	Caco-2 efflux; 3 folds; budget 100 features; random forest and LightGBM downstream	RMSE of the best single representation minus RMSE of the hybrid ($\log_{10}$ efflux ratio; positive means the hybrid has the lower error): DeepLIFT +0.0144, LightGBM gain +0.0100, TreeSHAP +0.0067, random ranking −0.0118	Full: LightGBM gain, TreeSHAP, DeepLIFT / fast: LightGBM gain
Internal evaluator settings	20 hyperparameter settings from 5 model families	Buchwald–Hartwig yield, Caco-2 efflux, Koenig expression; 3 folds; budget 100 features	Spread of mean scores across all 20 settings: 0.0253 R^2, 0.0179 R^2 and 0.0085 Spearman. Median standard deviation across the 3 folds for one setting, allocation method and downstream model: 0.0462 R^2, 0.0693 R^2 and 0.0186 Spearman	Matched, random forest, LightGBM and DNN, at fixed settings, both tiers

Setting	Values compared	Datasets, folds, budgets	What was measured, and the result	Values kept (full / fast)
Rescaling of each representation's score (a rule)	5 rescalings	9 datasets; 5 folds; budgets 50 and 300 features	Whether any rescaling beat the one in use by more than that rule's own spread across folds: in 0 of 90 (dataset, allocation method, budget) combinations	One fixed rule per metric (Chapter 5)

The representation-count and allocation rows come from one design run on 10 datasets, 3 downstream models and 2 budgets (60 combinations); QM9 is left out, which gives the 54.

Values of the search settings. Before the full runs, we ran a series of preliminary design experiments to determine the values each setting of the search may take. They used 1 to 9 datasets and 1 to 5 folds, and each dataset was scored in its own metric, with scores never averaged across datasets. When making a selection, we considered whether a value gave the best score for some dataset, downstream model and budget, as well as the compute time in hours. A value that won somewhere but cost many hours stayed in the full search and was left out of the fast tier. Each setting's values were then either kept, because each won for some dataset, model and budget, or fixed at one value. A setting was fixed when the spread between its values was smaller than the spread of one value across folds. As an example of a kept value, nine downstream models were fitted on each of 10 base representations over 5 folds. The datasets were Buchwald–Hartwig yield, Caco-2 efflux, Koenig expression, LogD and breast cancer PAM50 subtype. The DNN scored best on 4 of the 5 datasets and third on Caco-2 (0.2936 against 0.2851 RMSE, $\log_{10}$ efflux ratio), so it was added. As an example of a fixed rule, we tested five ways of rescaling each representation's score before the budget is split. On hERG K_i, with linear allocation at 300 features, the five rescalings gave RMSE values from 0.5870 to 0.5924 ($\mathrm{p}K_i$). A single rescaling, however, ranged from 0.5475 to 0.6294 RMSE across the 5 folds, so rescaling is one fixed rule per metric. Every design experiment is listed in Table S.15.

Importance methods. Two comparisons were run, the first of how the methods rank features and the second of the hybrids those rankings build. The ranking comparison scored 8 importance methods on 7 datasets, namely LogD, hERG

K_i, Buchwald–Hartwig yield, nanobody thermostability, antibody FcRn retention, breast invasive carcinoma (BRCA) PAM50 subtype and lung adenocarcinoma (LUAD) vs lung squamous cell carcinoma (LUSC). For each pair of methods, it measured the Spearman correlation between their scores over every feature of every representation. The three retained methods rank features differently. Across the 7 datasets, the correlation ranged from 0.20 to 0.64 between LightGBM gain and TreeSHAP, from 0.21 to 0.67 between LightGBM gain and DeepLIFT, and from 0.56 to 0.99 between TreeSHAP and DeepLIFT. These numbers are preliminary, because each method was fitted once on the whole dataset rather than inside folds.

The hybrid comparison used Caco-2, with 2,158 molecules, 10 representations, 3 scaffold folds and a budget of 100 features. Hybrids were built from each method's ranking and scored with random forest and LightGBM downstream models. Three control rankings were added, namely feature variance, Laplacian score and a random order. The measure was the RMSE of the best single representation minus the RMSE of the hybrid, in $\log_{10}$ efflux ratio units, so that a positive value means the hybrid has the lower error. The preliminary results were $+0.0144$ for DeepLIFT, $+0.0100$ for LightGBM gain, $+0.0067$ for TreeSHAP and -0.0118 for the random ranking. The gap between DeepLIFT and LightGBM gain, 0.0044 RMSE, is smaller than the fold-to-fold spread. The methods also differ at the top of their rankings. Of each representation's 100 top-ranked features on Caco-2, the retained pairs of methods share a median of 28 to 54 features.

Three rules determined which methods were retained. First, a method must score every feature of every representation for regression, binary and multiclass labels, which all three retained methods do. Second, of two methods whose rankings agree above a Spearman correlation of 0.95, we keep the faster one. Under this rule, DeepLIFT replaced integrated gradients, which agreed with it above 0.95 on 4 of 6 datasets, and gradient SHAP, which agreed above 0.95 on 3 of 6. Third, we keep methods that rank features differently, because a different ranking builds a different hybrid. TreeSHAP came last of the three on Caco-2 and was kept under this third rule. Only two filters act on features before allocation, the cut at a score of 0 and the pre-screen on representations wider than 2,000 features. On a 2,048-bit fingerprint, the pre-screen drops the 48 lowest-variance bits. On hERG K_i all 48 are among the 105 constant ECFP4 bits, while on QM9, which has no constant ECFP4 bits, they are near-constant ones. Near-constant features otherwise remain candidates and are ranked by importance.

No single importance method suits every dataset and downstream model. A preliminary run used 5 datasets (nanobody thermostability, hERG K_i, humanness, LogD and BRCA PAM50 subtype), 5 folds, 10 representations, 5 downstream models and budgets of 25, 50, 100, 150 and 300 features. For each of the 25 pairs of downstream model and budget per dataset, we counted which importance method gave the best configuration, and Table S.16 gives the counts. LightGBM gain won the most pairs on three datasets and tied with TreeSHAP on humanness. DeepLIFT won the most on one dataset, LogD, with 17 of its 25 pairs, and outside LogD it gave the best configuration for 2 of 100 pairs. With the DNN downstream model, LightGBM gain won 17 of 20 budgets outside LogD, whereas on LogD the DNN took DeepLIFT at 3 of 5 budgets. On LogD, XGBoost and LightGBM each took DeepLIFT at 5 of 5 budgets, and random forest took LightGBM gain at 5 of 5. **[RESULT NEEDED: the same win counts, per dataset and importance method, from the full runs]**

		Wins (of 25)			Best score		
Dataset	Metric	LightGBM gain	TreeSHAP	DeepLIFT	LightGBM gain	TreeSHAP	DeepLIFT
Nanobody thermostability	Spearman	15	10	0	0.6696	0.6471	0.6440
hERG K_i	RMSE (pK_i)	14	11	0	0.5589	0.5605	0.5728
Humanness	AUROC	12	12	1	0.9723	0.9713	0.9689
LogD	RMSE (log units)	8	0	17	0.4250	0.4345	0.4170
BRCA PAM50 subtype	weighted F1	21	3	1	0.9097	0.9018	0.9038

Table S.16: Importance method of the best configuration in a preliminary run on five datasets, with 5 folds, 10 representations, 5 downstream models and budgets of 25, 50, 100, 150 and 300 features. Wins count the pairs of downstream model and budget, out of 25 per dataset, at which a method gave the best configuration, and Best score is the best score reached with that method.

Allocation methods. We first tried greedy forward selection, which picks features for how well they work together, something fixed shares cannot do [10, 11]. In each round, it added the next block of features from whichever representation most improved the internal validation score. It stopped when the budget was spent or when the score had not improved for a set number of rounds. With blocks of s features and K representations, it needs up to $(B/s) \times K$ fits per configuration and fold. With blocks of 10 features, this amounts to 500 fits at 500 features and 10 representations, against 10 for the score-based methods. In preliminary runs, it reached a Spearman correlation of 0.3600 on Jain antibody FcRn retention, against 0.4526 for linear allocation, and placed 16th of 17 methods. On breast cancer PAM50

subtype, it reached an accuracy of 0.8551 against 0.8938 for linear allocation, again placing 16th of 17, and it was removed from the search.

The five methods were chosen in a preliminary comparison run before the main experiments. It compared 8 candidate allocation methods, each at 4 representation counts, on 9 regression datasets. It used 3 downstream models (LightGBM, random forest and CatBoost) and budgets of 500 and 1,000 features, giving 54 combinations of dataset, downstream model and budget. A fixed-share method was the single best method in 38 of the 54 combinations and the global pool in 16. All five were kept because no single method was best in most combinations. This comparison predates the current score rescaling, and it sized the weights with an inner 3-fold group split instead of the 80/20 internal split used now.

Matched internal evaluator. Matching the evaluator to the downstream model is the obvious default. However, the evaluator only has to rank representations, and another model may rank them better for the downstream model. Because the matched evaluator is one of the four searched values, the search already covers this default, and a mismatched evaluator winning is informative rather than a negative result. We first counted wins in the search itself, in a preliminary run on 5 datasets (nanobody thermostability, hERG K_i, humanness, LogD and breast cancer PAM50 subtype). These runs used 10 representations, budgets of 25 to 300 features and 5 folds, and their scores are the search's own fold scores rather than held-out scores. The matched evaluator gave the best score for its downstream model in 9 of 25 pairs of dataset and downstream model. Counted per budget, it was best in 43 of 125 combinations of dataset, downstream model and budget. With 3 or 4 evaluators per downstream model and no effect of matching, about 38 of 125 would be expected. On nanobody thermostability with a DNN downstream model, the matched evaluator reached a Spearman correlation of 0.6696 against 0.6506 for the LightGBM evaluator, a gap of 0.0190. Its 5 fold scores ranged from 0.5095 to 0.7920. On LogD with an SVM downstream model, the matched evaluator scored 0.4322 RMSE (log units) against 0.4217 for the LightGBM evaluator, where lower is better. This is a gap of 0.0105, the matched evaluator lost at all 5 budgets, and its 5 fold scores ranged from 0.4118 to 0.4634. We then scored the evaluators on held-out folds in a second preliminary run, with 3 outer folds, a budget of 250 features, 5 representations and LightGBM gain only. The downstream models were random forest and SVM. On nanobody thermostability with a random forest downstream model, the

matched evaluator reached a Spearman correlation of 0.6098, the LightGBM evaluator 0.6220 and the SVM evaluator 0.6172. The matched evaluator's fold scores ranged from 0.5659 to 0.6734. With an SVM downstream model, the matched evaluator reached 0.6189, the Gaussian process evaluator 0.6265 and the LightGBM evaluator 0.6317, with fold scores for the matched evaluator from 0.5530 to 0.6703. On hERG K_i with a random forest downstream model, the matched evaluator scored 0.6046 RMSE ($\mathrm{p}K_i$), the SVM evaluator 0.6059 and the LightGBM evaluator 0.6075. Its fold scores ranged from 0.5957 to 0.6178, and it beat the SVM evaluator on 2 of 3 folds. With an SVM downstream model, the matched evaluator scored 0.5723 RMSE ($\mathrm{p}K_i$), the LightGBM evaluator 0.5724 and the Gaussian process evaluator 0.5765, with fold scores for the matched evaluator from 0.5605 to 0.5862. Every gap above, both held out and within the search, is smaller than the range of the matched evaluator's own fold scores. **[RESULT NEEDED: per dataset and downstream model, how often the winning configuration used the matched evaluator, and its gap to the best other evaluator in the dataset's metric]**

Fast tier. The fast lists were chosen by one rule in a preliminary comparison. A value stays in the fast lists if it is inexpensive and improves the best score, whereas an expensive value that wins often stays in the full search only. Importance methods and downstream models were measured on five datasets, with 5 folds each, against the unpruned full search. The five were breast cancer PAM50 subtype, nanobody thermostability, humanness, hERG K_i and LogD. DeepLIFT took about half of the fast run's hours on all five datasets (49.9% to 51.6%), and dropping it changed only LogD's best score, from 0.4905 to 0.4930 RMSE (log units). Adding the DNN as a downstream model moved LogD from 0.4930 to 0.4273 RMSE, against a full-search best of 0.4170. Adding the SVM moved hERG K_i from 0.5776 to 0.5599 RMSE ($\mathrm{p}K_i$), while adding XGBoost changed no dataset's best score. All five downstream models are now searched at both tiers. The global pool won 60 of the 125 combinations of dataset, downstream model and budget on those five datasets. It is expensive, however, so it is in the full search only. The three fast allocation methods are linear, squared and rank-decay allocation. **[RESULT NEEDED: the share of run time the fast tier and adaptive pruning saved on the full runs]**

Additional file 16: Fields a blueprint saves

One row is one field. The first block is the recipe that rebuilds a hybrid; the second is kept for inspection and is not read by a rebuild.

Table S.17: Fields a blueprint saves. The stored score of each representation is its best score on its own across the downstream models, over every molecule the search ran on. It is not the internal-evaluator score that splits the budget, which a rebuild computes again on the molecules it is given. Two further fields are read only by allocation or augmentation routes outside the thesis's search and are fixed: a minimum improvement a whole block of extra features must show (none) and a number of features taken from each source (20).

Field	What it holds
The recipe	
Representations	The representations offered to the search
Representation count	How many of them the configuration used; empty when it used all
Allocation method	One of the five described in Chapter 5
Importance method	LightGBM gain, TreeSHAP or DeepLIFT
Budget	Number of features in the hybrid
Internal evaluator	Matched, random forest, LightGBM or DNN
Objective	Predictive, or predicting each molecule's activity from its five nearest neighbors
Metric	The dataset's metric, and whether lower is better
Task type	Regression or classification
Folds and seed	The number of folds, and the random seed
The recipe: augmentation	
Augmentation setting	None, uncapped or capped
Candidate ranking	How candidate extra features are ranked
Candidate list length	50 features
Cap per source	Optional; the most candidates taken from one source
Margin	How much better than adding none a block of extra features must score
Fraction of the budget	Optional; used instead of the fixed block sizes
Kept for inspection, not read by a rebuild	
Representation scores	Each representation's best score on its own across the downstream models, sign flipped for metrics where lower is better
Features per representation	The number of features each representation got in each fold
Extra features kept	The extra features augmentation kept in each fold, by source

Additional file 17: Reduction of wide representations when the comparator is chosen

A representation wider than 5,000 features is first reduced, inside each fold, to the 500 features on which a ranking model splits most often. It keeps 200 features instead when the search is given fewer than 1,000 molecules. The ranking model

is a LightGBM model of 50 trees of depth at most 5, fitted on the fold's training molecules. It ranks features by how many times its trees split on them rather than by gain. This reduction applies only when the search scores a single representation on its own.

Additional file 18: Worked example of fold counts

When a training fold holds fewer than five groups, both splits use as many folds as there are groups. For example, Koenig expression has five groups, so each outer training fold holds four. The search inside that fold uses 4 folds of three groups each, and the internal split holds out one of the three.

S.4 Hybrid representations for similarity and label noise

Additional file 19: Settings of the twelve LNL methods and the plain network they share

One row is one method; the first row is the plain network every method starts from. The kind of each method follows Chapter 6.

Table S.18: Settings of the twelve LNL methods. On classification datasets the squared-error loss is replaced by cross-entropy.

Method	Kind	Settings	Labels
Plain network	–	Hidden layers of 256, 128 and 64 units, ReLU, dropout 0.2; Adam at learning rate 0.001, halved after 5 epochs without improvement; batch size 256; gradient norm clipped at 1; up to 200 epochs; stops after 20 epochs without improvement on 20% of the training molecules held out by scaffold group (drawn at random after a method has removed molecules); labels z-normalized	both
Small-loss selection	noise mitigation	10 warm-up epochs on all molecules; keeps the 70% with the smallest loss and fits a new plain network on them	both
Confident learning	noise mitigation	3 plain networks; removes molecules whose error is above the 85th percentile and whose spread across the 3 networks is below the 25th percentile, then refits	both

Method	Kind	Settings	Labels
Removal by distance to neighbors	noise mitigation	10 nearest neighbors in the standardized features; score is the distance of a label from its neighbors' mean label, divided by their standard deviation (on classification, the share of neighbors with another label); removes molecules above the 85th percentile, then refits	both
Staged removal	noise mitigation	3 stages; each removes the 10% of the remaining molecules with the largest training loss	both
Per-epoch down-weighting	noise-robust training	After each epoch, the weight of every molecule whose loss is above the median is multiplied by 0.95	both
Curriculum by confidence	noise-robust training	20 warm-up epochs, then trains on the 25%, 50%, 75% and 100% of molecules with the smallest loss, in stages	both
DivideMix	noise-robust training	Two networks, 10 warm-up epochs; a two-component Gaussian mixture on each network's squared errors splits the molecules	both
Meta-Weight-Net	noise-robust training	The weighting network is updated every 5 epochs at learning rate 0.001, on 20% of the training molecules held out for it	both
MentorNet	noise-robust training	The mentor network reads each molecule's current loss, its last 5 losses and its mean distance to its 10 nearest neighbors	both
Contrast to Divide	noise-robust training	5 perturbed copies of each molecule, Gaussian noise of standard deviation 0.01 on the features	both
Heteroscedastic GP	noise-robust training	RBF kernel, 2 optimizer restarts, fitted on at most 1,000 training molecules; a small network predicts each molecule's noise variance	regression
Deep evidential regression	noise-robust training	Weight of the evidence penalty 0.01	regression

S.5 Datasets and curation

This appendix gives, for every dataset, its source, its label and each curation step in the order the steps run. Chapter 5 and Table 5.8 give the labels, metrics, fold groupings and splitting rules of its 16 datasets. Table S.19 gives the sample counts before and after curation, obtained by running the curation code on the downloaded files.

Dataset	Released	Used	Steps that remove samples
Small molecules			
hERG K_i	1,815	1,412	no pChEMBL value (3); not a binding assay (75); repeats merged into one median per compound (322); InChIKey duplicates (3)
LogD	5,326	5,028	no value (287); repeats merged after standardization (11)
Caco-2 efflux	5,326	2,158	no value (3,165); repeats merged after standardization (3)
Rat liver microsome clearance	3,521	3,053	no value (467); InChIKey duplicates (1)
MDR1–MDCK efflux	3,521	2,641	no value (879); InChIKey duplicates (1)
QM9	130,831	9,978	not parsable by RDKit (1,403); random draw of 10,000; InChIKey duplicates (22)
hERG blockers, training set	7,772	7,606	also in the hERG K_i set (166)
hERG blockers, test set	4,383	4,213	also in the hERG K_i set (170)
hERG IC_{50}	12,010	12,008	missing or non-positive value (2)
Chemical reactions			
Buchwald–Hartwig yield	3,955	3,955	none
Buchwald–Hartwig plates	4,599	4,312	no aryl halide (287)
Antibodies			
Koenig expression	4,275	4,275	none
Nanobody thermostability	672	672	none
Humanness	564	564	none
Trastuzumab–HER2 binding	524,346	5,000	stratified random draw of 5,000
Nucleotide sequences			
H3K4me3	18,244	18,244	none (training and test files merged)
TATA promoter	5,274	5,274	none (training and test files merged)
Cancer genomics (TCGA)			
Lung adenocarcinoma vs squamous	1,129	1,017	not a primary tumor (112)
Breast cancer PAM50 subtype	1,097	821	no PAM50 call, or called normal-like (276); count is of primary tumors

Table S.19: Sample counts of each dataset before and after curation. Released is the number of rows in the downloaded files (for breast cancer, of primary tumors with expression data), and Used the number left after curation. The last column lists the steps that remove samples, with the number removed where it applies.

S.5.1 Small Molecules

Three steps are shared by several small-molecule datasets. The first is structure standardization with RDKit [12]: a general cleanup, keeping the largest fragment, neutralizing charges and selecting the canonical tautomer, in that order. Molecules that then contain a metal atom are rejected. Standardization is applied to hERG K_i, LogD and Caco-2; the Biogen sets, QM9 and the hERG blocker sets are only checked to parse in RDKit.

The second removes duplicate molecules by InChIKey [13], keeping the first occurrence, so that one molecule cannot sit in both a training and a test fold. It is applied to hERG K_i, LogD, Caco-2, the two Biogen sets and QM9, and to the hERG blocker sets only across sites (below). The third is fold grouping by Bemis–Murcko scaffold [14] (Chapter 5).

hERG K_i. hERG records (ChEMBL target CHEMBL240) were retrieved from ChEMBL [15], keeping binding assays (assay type B) with standard type K_i, a pChEMBL value, an exact relation ("=") and no curator data-validity comment. The label is the pChEMBL value, $\mathrm{p}K_i = -\log_{10} K_i$ with K_i in molar units. Repeat measurements of one compound are merged into their median, and compounds whose repeats have a standard deviation above 1.0 $\mathrm{p}K_i$ unit are removed, following Landrum and Riniker [16]. This gives 1,415 compounds, 226 of them measured more than once and none removed by the standard-deviation rule. After standardization, InChIKey deduplication removes 3, leaving 1,412 compounds with $\mathrm{p}K_i$ from 4.07 to 9.41. Because repeats are merged on ChEMBL's own SMILES before standardization, two ChEMBL entries that standardize to the same molecule are not merged by median; the InChIKey step keeps the first of them. The file read by the experiments was pulled on 1 February 2026; the ChEMBL release was not recorded at the time. Re-running the same query in September 2026 against ChEMBL release 37 gives a byte-identical file of 1,415 compounds.

LogD and Caco-2 efflux. Both come from the training split of the OpenADMET–ExpansionRx challenge data on Hugging Face [17]. For each endpoint, rows without a value are dropped, structures are standardized, and rows that standardize to the same SMILES are merged into their median label. The unitless efflux ratio is transformed to $\log_{10}$. Labels are clipped at the 1st and 99th percentiles of each dataset, -1.0 and 4.5 for LogD and -0.319 and 1.679 for the

$\log_{10}$ efflux ratio. This sets 116 LogD and 45 Caco-2 labels to a limit before any noise is added in Chapter 6. InChIKey deduplication removes nothing.

Rat liver microsome clearance and MDR1–MDCK efflux. These are two endpoints of the Biogen ADME set of 3,521 compounds [18], from Polaris as `biogen/adme-fang-v1` [19]: intrinsic clearance $\mathrm{CL}_{\mathrm{int}}$ in mL/min/kg and an efflux ratio, both released and used as $\log_{10}$ values. Rows without a value, or whose SMILES RDKit cannot parse, are dropped, and InChIKey deduplication removes one molecule from each. There is no standardization, median merge or clipping.

QM9. QM9 [20, 21] was read through PyTorch Geometric [22], which distributes 130,831 molecules. Of these, 129,428 parse in RDKit. From these, 10,000 molecules were drawn once at random with seed 42, so the subsample is a random draw of molecules, not a random split. InChIKey deduplication removes 22, leaving 9,978. The label is the HOMO–LUMO gap in eV, from 1.88 to 11.37 eV in this sample.

hERG blockers held at separate sites. The separate-site experiments of Chapter 5 use two further hERG sets from Lhasa Limited [23, 24]: a training set of 7,772 molecules (3,630 blockers, 46.7%) and a test set of 4,383 molecules (1,084 blockers). Each label is Lhasa's own binary blocker call, made under different rules in the two sets. Both are proprietary (Additional file S.6). The hERG K_i set acts as a third site, with a compound counted as a blocker when $\mathrm{p}K_i \geq 6.0$, giving 467 blockers and 945 non-blockers. Molecules are matched by InChIKey across the sites, in the order hERG K_i, training set, test set, keeping the first copy. This removes 166 training and 170 test molecules, all of them also in the hERG K_i set; the two Lhasa sets share none.

hERG IC_{50}. This set is used only in Chapter 6, where its repeat measurements and curator flags serve as real evidence of label error. It keeps every measurement as its own row. The query takes ChEMBL records for CHEMBL240 with standard type IC_{50} in nM and an exact relation [15], dropping records with a missing or non-positive value. The label is $\mathrm{pIC}_{50} = 9 - \log_{10}(\mathrm{IC}_{50}/\mathrm{nM})$. Assay type is not filtered. Of the 12,010 records, 10,219 come from binding assays, 1,543 from toxicity assays, 217 from functional assays and 31 from ADME assays, as ChEMBL classifies them. The 301 records with a curator data-validity comment, on 215 molecules, are kept

and marked. With no standardization, median merge or InChIKey step, the result is 12,008 measurements of 9,802 molecules. Folds are grouped by scaffold, so no molecule is split between training and test.

S.5.2 The Chapter 3 versions of LogD, Caco-2, hERG K_i and QM9

Chapter 3 used the same sources as Chapter 5 for LogD, Caco-2 and hERG K_i but a lighter curation, which explains the different sample counts (Table S.20). Its QM9 replicates are drawn from the same 129,428 parsable molecules (Section 3.2). It also merges hERG K_i repeats a second time, on its own standardized SMILES; this merges nothing, so its set keeps 1,415 compounds.

	Chapter 3	Chapter 5
Structure standardization	largest fragment and canonical SMILES only	cleanup, largest fragment, neutralized charges, canonical tautomer, metals rejected
Label clipping (LogD, Caco-2)	none	1st and 99th percentile
InChIKey deduplication	none	first occurrence kept
Scaffold key	stereochemistry removed; each acyclic molecule its own group	stereochemistry kept; all acyclic molecules one group
Molecules: LogD, Caco-2, hERG K_i	5,039; 2,161; 1,415	5,028; 2,158; 1,412

Table S.20: Curation of LogD, Caco-2 and hERG K_i in Chapters 3 and 5. The last row gives the numbers of molecules.

S.5.3 Chemical Reactions

Both reaction datasets come from the high-throughput Buchwald–Hartwig C–N coupling experiments of Ahneman et al. [25], which varied aryl halide, palladium ligand, base and additive. The label is the yield in %, from 0 to 100. One sample is a whole reaction, written as its reactants and product.

Buchwald–Hartwig yield. This is the version released with Schwaller et al. [26], from that paper's public repository: 3,955 reactions. Each is written as aryl halide, 4-methylaniline, ligand, base and additive going to a product, generated from the

aryl halide by a C–N coupling template that succeeds on all 3,955. They fall into 22 additive groups of 178 to 180 reactions.

Buchwald–Hartwig plates. This is the data table of the public repository of Ahneman et al. [25]: 4,599 rows over three plates, with product SMILES from the file. The 287 rows without an aryl halide are dropped. The 180 remaining rows without an additive form their own group, giving 24 additive groups of 177 to 180 reactions. The 4,312 reactions kept include all 3,955 of the first set, with the same aryl halide, ligand, base, additive and yield, plus 357 more: the 180 without an additive and 177 with 5-phenyl-1,2,4-oxadiazole as the additive.

S.5.4 Antibodies

The Koenig and nanobody sets were downloaded as CSV files from the FLAb benchmark repository [27], the humanness set from the BioPhi publication repository and the trastuzumab set from its own repository. The FLAb and BioPhi files come from each repository's main branch, with no commit pinned. Each file has been changed only once. The Koenig and nanobody files were changed in FLAb commits `c41e56e` and `005e6c1`, both on 6 August 2025, and the two humanness files in BioPhi commit `c0797f2` on 24 August 2021. The local copies of the Koenig and humanness files are identical to those commits.

The nanobody and humanness sets are grouped by sequence identity, defined for sequences a and b as

$$\mathrm{id}(a, b) = 1 - \frac{\mathrm{lev}(a, b)}{\max(|a|, |b|)}, \tag{S.1}$$

where lev is the Levenshtein (edit) distance and $|a|$ is the length in residues. Following Kenlay et al. [28], sequences are clustered by single linkage at id ≥ 0.95. The 672 nanobodies fall into 309 clusters and the 564 humanness antibodies into 461.

Koenig expression. This deep mutational scan of the anti-VEGF antibody G6 [29], taken from FLAb, has 4,275 variants with both chains (heavy chains of 120 residues, light chains of 108). Each carries exactly one mutation, 2,261 in the heavy chain and 2,014 in the light chain, at 225 distinct positions. The label is the unitless expression enrichment ratio, from 0.00098 to 3.178.

Nanobody thermostability. This is the NbThermo set [30], taken from FLAb: melting temperatures T_m from 39.0 to 98.2 °C for 672 single-domain antibodies with no light chain. The 672 rows hold 556 distinct sequences; 116 appear twice, and 106 of those pairs carry different melting temperatures, up to 8.8 °C apart. These repeats are kept, so the labels carry disagreement between measurements before any noise is added.

Humanness. This is the list of 564 therapeutic antibodies from IMGT/mAb-DB released with BioPhi [31], using the first sequence listed for each. An antibody is labeled human (1) only when its species field is exactly "Human" (198 antibodies). The other 366 are labeled non-human (0). The 564 antibodies hold 506 distinct sequences, mostly because an antibody and its drug conjugate share a sequence. One sequence is shared by three antibodies with different labels, two humanized and one human. All three sit in one identity cluster, so they never fall on both sides of a split.

Trastuzumab–HER2 binding. This is the library of trastuzumab variants in the ten-residue CDR-H3 (the third complementarity-determining region of the heavy chain), screened for HER2 binding by Chinery et al. [32]. The released training, validation and test files are joined and re-split. All 524,346 variants have a 10-residue CDR-H3 and are kept; 32.8% are binders. From these, 5,000 variants were drawn once with seed 42, stratified so that each combination of label and number of mutated CDR-H3 residues keeps its share; the sample holds 1,639 binders (32.8%). Each 120-residue heavy chain is rebuilt by inserting the variant CDR-H3 into the trastuzumab framework, and paired with the fixed trastuzumab light chain.

S.5.5 Nucleotide Sequences

Both nucleotide tasks come from the revised downstream tasks of the Nucleotide Transformer [33] on Hugging Face (`InstaDeepAI/nucleotide_transformer_downstream_tasks_revised`). The download is not pinned to a revision; the files used are those of revision `851f994`, dated 30 June 2025, the latest at the time of writing. The released training and test files are merged, and folds are grouped by the chromosome read from each sequence's name. Nothing else is filtered. Both tasks cover chromosomes 1 to 21, X and Y; chromosome 22 is absent.

H3K4me3. Each sample is a 1,000-base sequence, labeled by whether it carries the H3K4me3 histone mark. The merged files hold 18,244 sequences, 9,122 per class. The dataset card lists 30,000 training sequences; the count here is that of the files.

TATA promoter. Each sample is a 300-base sequence, labeled by whether it is a promoter with a TATA box. The merged files hold 5,274 sequences, 2,637 per class.

S.5.6 Cancer Genomics (TCGA)

The two cancer datasets use TCGA data [34] from UCSC Xena [35]. Expression is each cohort's Illumina HiSeq RNA-seq matrix on the Xena TCGA hub (version 2017-10-13): the same 20,530 genes in every cohort, unfiltered, as $\log_2(x + 1)$ of the RSEM-normalized count [36]. Only primary tumors are kept, one sample per patient. Missing expression values are set to 0. No batch correction is applied, because every fold holds out whole tissue source sites and a correction fitted on the training sites cannot adjust an unseen site, and a correction fitted on all sites would already have seen the test samples.

Lung adenocarcinoma against lung squamous cell carcinoma. The adenocarcinoma [37] and squamous cell carcinoma [38] cohorts are stacked, with the cohort as the label. Dropping 110 normal-tissue and 2 recurrent-tumor samples leaves 515 adenocarcinomas and 502 squamous tumors, 1,017 in all, from 68 tissue source sites.

Breast cancer PAM50 subtype. The label is the PAM50 subtype call [39] from the Xena breast cancer clinical matrix [40]. Tumors called normal-like or without a call are dropped, leaving 821 of 1,097 primary tumors with expression data: 421 luminal A, 192 luminal B, 141 basal-like and 67 HER2-enriched, from 19 tissue source sites.

S.6 Data and code availability

The software behind this thesis includes NoiseInject, the label-noise package released with Chapter 3, and the KIRBy repository, which implements the adaptive hybrid representation and holds the experiments of Chapters 4, 5 and 6 and the assay-dataset runs of Chapter 3.

NoiseInject is available as an open-source Python package under an MIT license [41]. Its noise conditions and metrics are described in Section 2.5.1.

- Project name: NoiseInject
- Project home page: https://github.com/adpunt/noise_inject
- Archived version: https://doi.org/10.5281/zenodo.20532710
- Version: 1.0.0
- Operating system(s): Platform independent
- Programming language: Python 3.7 or later
- Other requirements: NumPy, pandas, SciPy and scikit-learn; see the repository's `requirements.txt`
- License: MIT
- Any restrictions to use by non-academics: None

The QM9 results of Chapter 3, with that chapter's hyperparameter tuning, variance decomposition and figure scripts, come from a separate repository in Python and Rust, described in Chapter 3 under "QM9 Data Pipeline".

- Project name: qsar_qm_models
- Project home page: https://github.com/adpunt/qsar_qm_models
- Archived version: not yet archived
- Operating system(s): Linux and macOS
- Programming language: Python and Rust (2021 edition)
- Other requirements: RDKit installed through conda, to which the Rust program links; PyTorch and PyTorch Geometric
- License: GNU GPL v3.0
- Any restrictions to use by non-academics: those of the GNU GPL v3.0

KIRBy is written in Python and works on feature matrices computed beforehand, so the hybrid search applies to representations from any domain. It holds the hybrid search, loaders for the datasets in Additional file S.5, generators for the representations, the similarity analyses and learning-with-noisy-labels methods of Chapter 6, and every experiment script. Label noise is added through NoiseInject, so the noise definitions match Chapter 3.

- Project name: KIRBy
- Project home page: `https://github.com/adpunt/KIRBy`
- Archived version: [Zenodo DOI]
- Operating system(s): Linux and macOS
- Programming language: Python 3.8 or later; several pretrained encoders need Python 3.10 or later
- Other requirements: NumPy, pandas, SciPy, scikit-learn, and RDKit installed through conda; each application area has an optional install
- License: [license]
- Any restrictions to use by non-academics: [depends on the license]

All datasets but one are public. QM9 [20] is deposited on Figshare [21]. The LogD and Caco-2 efflux data are from the OpenADMET–ExpansionRx release under CC-BY-4.0 [17]. The hERG K_i data come from ChEMBL [15], released under CC BY-SA 3.0, and can be regenerated with the protocol in Additional file S.5. The rat liver microsome clearance and MDR1–MDCK efflux data are the Biogen ADME set [18], on Polaris [19] under CC-BY-4.0. The reaction-yield, antibody, nucleotide and tumor gene-expression data come from the public releases cited in Additional file S.5. Pretrained model weights come from their original sources and are not redistributed.

The exception is Lhasa Limited's curated hERG training and test sets [42] (Additional file S.5), used for feature selection with data held at separate sites (Chapter 5). They are proprietary and not redistributed. The comparison with FLuID also cannot be run without Lhasa's FLuID code. Every other result in this thesis can be reproduced from public data and the code above.

Lhasa Limited supplied the curated hERG data and the FLuID code [42]. The use of generative AI is described in the declaration in the front matter.

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